

```
In [5]: X = pd.get_dummies(X,dtype=int)

In [6]: from sklearn.model_selection import train_test_split
...: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 0)
...:

In [7]: from sklearn.linear_model import LinearRegression
...: regressor = LinearRegression()
...: regressor.fit(X_train,y_train)
Out[7]: LinearRegression()

In [8]: y_pred = regressor.predict(X_test)

In [9]: m = regressor.coef_
...: print(m)
[ 7.73467193e-01  3.28845975e-02  3.66100259e-02  8.66383692e+01
 -8.72645791e+02  7.86007422e+02]

In [10]: c = regressor.intercept_
...: print(c)
...:
42467.52924855314
```

```
[8]: y_pred = regressor.predict(X_test)

[9]: m = regressor.coef_
...: print(m)
.73467193e-01  3.28845975e-02  3.66100259e-02  8.66383692e+01
.72645791e+02  7.86007422e+02]

[10]: c = regressor.intercept_
...: print(c)
...:
67.52924855314

[11]: X = np.append(arr = np.full((50, 1), 42467).astype(int), values = X, axis = 1)

[12]: import statsmodels.api as sm
...: X_opt = X[:,[0,1,2,3,4,5]]
...: #OrdinaryLeastSquares
...: regressor_OLS = sm.OLS(endog=y, exog=X_opt).fit()
...: regressor_OLS.summary()
...:
```

```
Out[12]:
<class 'statsmodels.iolib.summary.Summary'>
"""
```

```

                OLS Regression Results
=====
Dep. Variable:          Profit      R-squared:          0.951
Model:                  OLS        Adj. R-squared:       0.945
Method:                 Least Squares  F-statistic:       169.9
Date:                  Fri, 31 Oct 2025  Prob (F-statistic): 1.34e-27
Time:                  00:10:51      Log-Likelihood:    -525.38
No. Observations:      50          AIC:                1063.
Df Residuals:          44          BIC:                1074.
Df Model:              5
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	1.1794	0.164	7.204	0.000	0.849	1.509
x1	0.8060	0.046	17.369	0.000	0.712	0.900
x2	-0.0270	0.052	-0.517	0.608	-0.132	0.078
x3	0.0270	0.017	1.574	0.123	-0.008	0.062
x4	41.8870	3256.039	0.013	0.990	-6520.229	6604.003

Console 1/A ×

```

Covariance Type:       nonrobust
=====
                coef    std err          t      P>|t|      [0.025    0.975]
-----
const          1.1794      0.164       7.204      0.000        0.849     1.509
x1              0.8060      0.046      17.369      0.000        0.712     0.900
x2             -0.0270      0.052       -0.517      0.608       -0.132     0.078
x3              0.0270      0.017       1.574      0.123       -0.008     0.062
x4             41.8870    3256.039       0.013      0.990    -6520.229    6604.003
x5             240.6758    3338.857       0.072      0.943    -6488.349    6969.701
=====
Omnibus:                14.782    Durbin-Watson:           1.283
Prob(Omnibus):           0.001    Jarque-Bera (JB):        21.266
Skew:                   -0.948    Prob(JB):               2.41e-05
Kurtosis:                5.572    Cond. No.               8.45e+05
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 8.45e+05. This might indicate that there are strong multicollinearity or other numerical problems.

"""

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```
In [13]: import statsmodels.api as sm
...: X_opt = X[:,[0,1,2,3]]
...: #OrdinaryLeastSquares
...: regressor_OLS = sm.OLS(endog=y, exog=X_opt).fit()
...: regressor_OLS.summary()
```

Out[13]:

```
<class 'statsmodels.iolib.summary.Summary'>
"""
```

OLS Regression Results

```
=====
Dep. Variable:          Profit    R-squared:                0.951
Model:                  OLS       Adj. R-squared:           0.948
Method:                 Least Squares   F-statistic:            296.0
Date:                  Fri, 31 Oct 2025   Prob (F-statistic):      4.53e-30
Time:                  00:14:11    Log-Likelihood:         -525.39
No. Observations:      50          AIC:                   1059.
Df Residuals:          46          BIC:                   1066.
Df Model:              3
Covariance Type:       nonrobust
=====
```

```
=====
coef    std err          t      P>|t|      [0.025      0.975]
-----
const    1.1803      0.155      7.626    0.000      0.869      1.492
x1        0.8057      0.045     17.846    0.000      0.715      0.897
x2       -0.0268      0.051     -0.526    0.602     -0.130      0.076
x3        0.0272      0.016      1.655    0.105     -0.006      0.060
=====
```

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Covariance Type: nonrobust

```
=====
coef    std err          t      P>|t|      [0.025      0.975]
-----
const    1.1803      0.155      7.626    0.000      0.869      1.492
x1        0.8057      0.045     17.846    0.000      0.715      0.897
x2       -0.0268      0.051     -0.526    0.602     -0.130      0.076
x3        0.0272      0.016      1.655    0.105     -0.006      0.060
=====
Omnibus:                 14.838    Durbin-Watson:           1.282
Prob(Omnibus):            0.001    Jarque-Bera (JB):        21.442
Skew:                     -0.949    Prob(JB):                2.21e-05
Kurtosis:                 5.586    Cond. No.                35.0
=====
```

Notes:

```
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
"""
```

```
In [14]: import statsmodels.api as sm
...: X_opt = X[:,[0,1,3]]
...: #OrdinaryLeastSquares
...: regressor_OLS = sm.OLS(endog=y, exog=X_opt).fit()
...: regressor_OLS.summary()
...:
```

```
Out[14]:
<class 'statsmodels.iolib.summary.Summary'>
"""
```

```

                        OLS Regression Results
=====
Dep. Variable:          Profit      R-squared:          0.950
Model:                  OLS         Adj. R-squared:       0.948
Method:                 Least Squares   F-statistic:        450.8
Date:                   Fri, 31 Oct 2025   Prob (F-statistic):  2.16e-31
Time:                   00:15:05         Log-Likelihood:     -525.54
No. Observations:       50             AIC:               1057.
Df Residuals:           47             BIC:               1063.
Df Model:                2
Covariance Type:        nonrobust
=====
```

```

Df Model:                2
Covariance Type:        nonrobust
=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const          1.1062      0.063     17.464      0.000      0.979      1.234
x1             0.7966      0.041     19.266      0.000      0.713      0.880
x2             0.0299      0.016      1.927      0.060     -0.001      0.061
=====
Omnibus:            14.677   Durbin-Watson:           1.257
Prob(Omnibus):       0.001   Jarque-Bera (JB):        21.161
Skew:                -0.939   Prob(JB):               2.54e-05
Kurtosis:            5.575   Cond. No.                13.0
=====
```

```
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
"""
```

```
In [15]: import statsmodels.api as sm
...: X_opt = X[:,[0,1]]
...: #OrdinaryLeastSquares
...: regressor_OLS = sm.OLS(endog=y, exog=X_opt).fit()
...: regressor_OLS.summary()
```

```
Out[15]:
<class 'statsmodels.iolib.summary.Summary'>
"""
```

OLS Regression Results

```
=====
Dep. Variable:          Profit    R-squared:                0.947
Model:                  OLS       Adj. R-squared:           0.945
Method:                 Least Squares   F-statistic:             849.8
Date:                   Fri, 31 Oct 2025   Prob (F-statistic):      3.50e-32
Time:                   00:15:29   Log-Likelihood:          -527.44
No. Observations:       50       AIC:                     1059.
Df Residuals:           48       BIC:                     1063.
Df Model:                1
Covariance Type:        nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
--	------	---------	---	------	--------	--------

```
Out Model:          1
Covariance Type:    nonrobust
```

```
=====
              coef    std err          t      P>|t|      [0.025    0.975]
-----
const         1.1546     0.060     19.320     0.000     1.034     1.275
x1             0.8543     0.029     29.151     0.000     0.795     0.913
=====
Omnibus:                 13.727   Durbin-Watson:           1.116
Prob(Omnibus):            0.001   Jarque-Bera (JB):         18.536
Skew:                    -0.911   Prob(JB):                 9.44e-05
Kurtosis:                 5.361   Cond. No.                  4.60
=====
```

Notes:

```
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
"""
```

```
In [16]: bias = regressor.score(X_train, y_train)
...: bias
```

```
Out[16]: 0.9501847627493606
```

Notes:

[1] Standard Errors assume that the covariance matrix of the e
"""

```
In [16]: bias = regressor.score(X_train, y_train)
...: bias
```

```
Out[16]: 0.9501847627493606
```

```
In [17]: variance = regressor.score(X_test, y_test)
...: variance
```

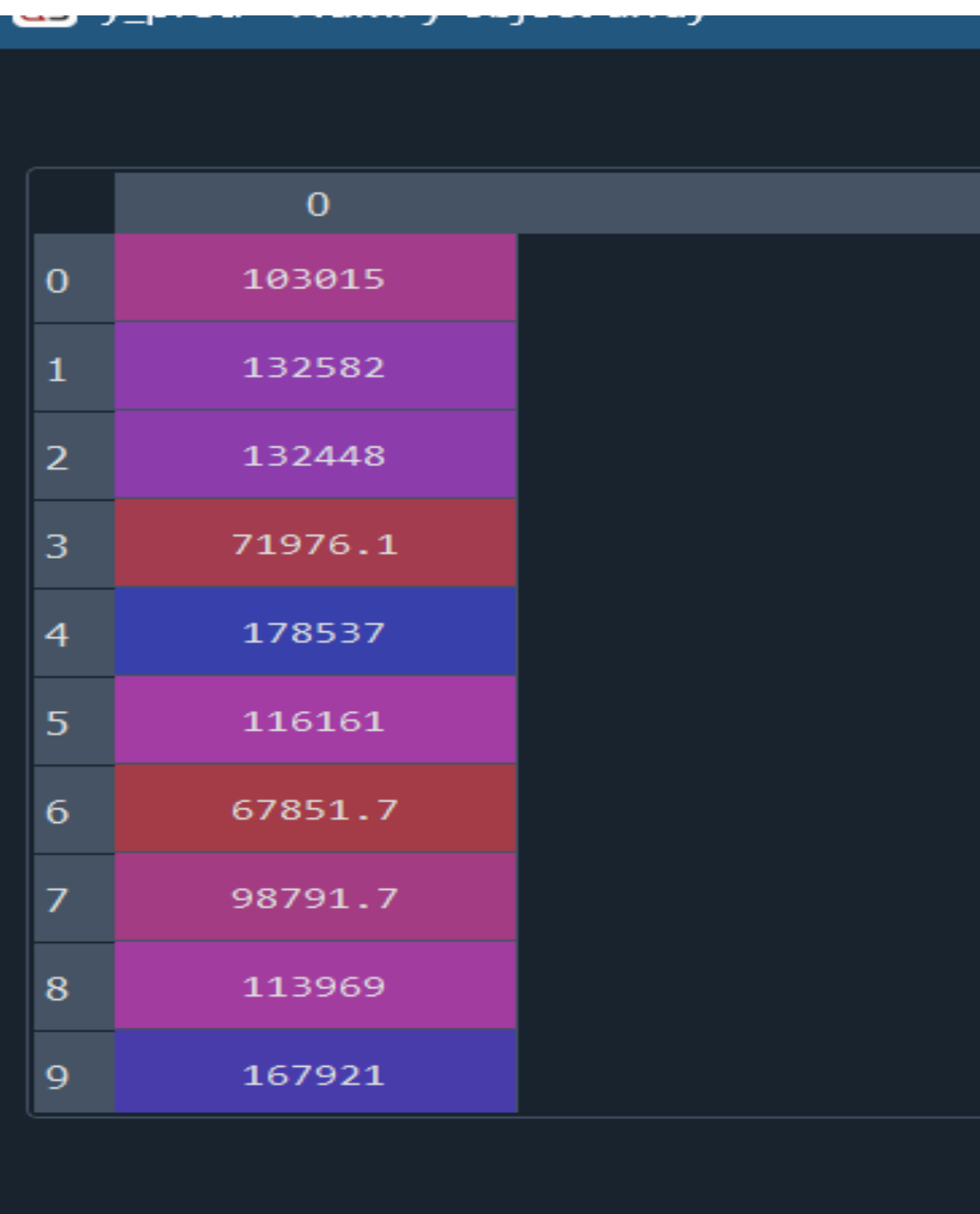
```
Out[17]: 0.9347068473282423
```

```
In [18]:
```



	Index	jitalMarketi	Promotion	Research	State	Profit
	0	165349	136898	471784	Hyderabad	192262
	1	162598	151378	443899	Bangalore	191792
	2	153442	101146	407935	Chennai	191050
	3	144372	118672	383200	Hyderabad	182902
	4	142107	91391.8	366168	Chennai	166188
	5	131877	99814.7	362861	Hyderabad	156991
	6	134615	147199	127717	Bangalore	156123
	7	130298	145530	323877	Chennai	155753
	8	120543	148719	311613	Hyderabad	152212
	9	123335	108679	304982	Bangalore	149760
	10	101913	110594	229161	Chennai	146122
	11	100672	91790.6	249745	Bangalore	144259
	12	93863.8	127320	249839	Chennai	141586
	13	91992.4	135495	252665	Bangalore	134307
	14	119943	156547	256513	Chennai	132603
	15	114524	122617	261776	Hyderabad	129917
	16	78013.1	121598	264346	Bangalore	126993
	17	94657.2	145078	282574	Hyderabad	125370
	18	91749.2	114176	294920	Chennai	124267
	19	86419.7	153514	0	Hyderabad	122777
	20	76253.0	143867	288664	Bangalore	118474

	0	1	2	3	4	5	6	
0	42467	165349	136898	471784	0	0	1	
1	42467	162598	151378	443899	1	0	0	
2	42467	153442	101146	407935	0	1	0	
3	42467	144372	118672	383200	0	0	1	
4	42467	142107	91391.8	366168	0	1	0	
5	42467	131877	99814.7	362861	0	0	1	
6	42467	134615	147199	127717	1	0	0	
7	42467	130298	145530	323877	0	1	0	
8	42467	120543	148719	311613	0	0	1	
9	42467	123335	108679	304982	1	0	0	
10	42467	101913	110594	229161	0	1	0	
11	42467	100672	91790.6	249745	1	0	0	
12	42467	93863.8	127320	249839	0	1	0	
13	42467	91992.4	135495	252665	1	0	0	
14	42467	119943	156547	256513	0	1	0	
15	42467	114524	122617	261776	0	0	1	
16	42467	78013.1	121598	264346	1	0	0	
17	42467	94657.2	145078	282574	0	0	1	
18	42467	91749.2	114176	294920	0	1	0	
19	42467	86419.7	153514	0	0	0	1	
20	42467	76253.9	113867	298664	1	0	0	



X_test	DataFrame	[10, 6]	Column names: DigitalMarketing, Promotion, Research, State_Bangalore, ...
X_train	DataFrame	[40, 6]	Column names: DigitalMarketing, Promotion, Research, State_Bangalore, ...
y	Series	[50]	Series object of pandas.core.series module
y_pred	Array of float64	[10]	[103015.20159796 132582.27760816 132447.73845174 71976.09851258 1785 ...
y_test	Series	[10]	Series object of pandas.core.series module
y_train	Series	[40]	Series object of pandas.core.series module

dataset	DataFrame	[50, 5]	Column names: DigitalMarketing, Promotion, Research, State, Profit
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